

CONG CHI TAI NGUYEN (JOSEPH)

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PROFESSIONAL SUMMARY

Experienced Engineer with over 10 years in software development, data engineering, and data science. Specialized in building and deploying end-to-end ML solutions, including real-time inference, model training, feature pipelines and skilled in Python, SQL. Recent work includes developing production-ready ML systems for geospatial prediction and supporting large-scale ETL pipelines. Recently graduated Postgraduate Certificates in Artificial Intelligence & Data Analytics and Cloud Computing at Saskatchewan Polytechnic. Passionate about responsible AI, risk modeling, and cross-functional collaboration.

TECHNICAL SKILLS

ML/AI: Random Forest, ResNet, Neural Networks, XGBoost, Model Deployment, Data Preprocessing

Programming: Python, .NET, SQL

Tools & Frameworks: scikit-learn, pandas, NumPy, Gradio, Docker, Kafka, Git, , RESTful APIs

Databases: PostgreSQL (with PostGIS), MySQL, MS SQL Server, Oracle, MongoDB, Cassandra

Cloud Platforms: Azure, AWS, GCP

Methodologies: Agile, Waterfall, SDLC

Other: Data Visualization, Feature Engineering, Model Monitoring, ETL

Soft Skills: Customer Service, Communication, Problem-Solving, Self-Learning, Team Collaboration

PROFESSIONAL EXPERIENCE

Research Assistant - Machine Learning

June 2024 - Present

Digital Integration Centre of Excellence (DICE)

- Built and deployed a Gradio-based ML inference web app with real-time predictions using satellite data
- Developed and trained ML models (XGBoost, Random Forest, ResNet, Neural Network) to classify geospatial patterns emphasizing feature engineering and calibration
- Supported end-to-end data pipelines using Python, SQL, and PostgreSQL; contributed to scalable model training and monitoring
- Conducted data validation and quality checks, and delivered insights to stakeholders through reports and visualizations

Freelancer Data Engineer

January 2024 - March 2024

TMAM Project (Part-time Remote)

- Designed and deployed a Python-based web crawler for structured data ingestion
- Built ETL pipelines to store and analyze data using PostgreSQL and MongoDB, enabling future ML model training

Senior Data Engineer

April 2022 - July 2023

Geniebook

- Implemented real-time data pipelines using Kafka, Python, and REST APIs to handle structured data
- Migrated relational datasets from MySQL to PostgreSQL, improving scalability for downstream analytics and AI models
- Led integration between internal data systems and Odoo ERP for actionable reporting
- Automated processes using Google Forms and Apps Script, improving data collecting efficiency

Experienced Data Consultant

August 2021 - April 2022

Amaris Consulting

- Translated business logic into optimized SQL and PL/SQL for reporting and model features
- Enhanced legacy data migration workflows, tuned complex SQL queries, and applied indexing for performance boosts
- Collaborated with cross-functional teams to deploy database-driven applications

Senior Database Developer

September 2020 - August 2021

Bolloré Transport & Logistics

- Built and maintained stored procedures, functions, and performance-optimized scripts
- Designed schema and pipelines for reporting on logistics KPIs using SQL Server and Power BI
- Collaborated with business teams to ensure usability of transformed data

Senior Officer – Core Banking IT

April 2014 - September 2020

Mizuho Bank

- Supported development of SQL-based financial reporting and internal data systems
- Maintained core banking applications and operational databases
- Built ETL pipelines using SSIS and maintained dashboards using Power BI
- Contributed to RPA process automation using UiPath for banking workflows
- Handled Excel/CSV-based data extractions, transformations, and uploads
- Delivered user documentation and technical support for banking systems

Senior Software Developer

December 2013 - March 2014

UNIT Technology Corporation

- Implemented internal HR systems using C#, ASP.NET MVC, and SQL Server, including backend database design and user interface development
- Built PowerShell modules to collect system data for remote monitoring software

Programmer

September 2009 - May 2013

FPT Information System

- Designed, developed, and deployed web applications in ASP.NET and SQL-based reports for enterprise clients
- Participated in application delivery lifecycle, including QA and deployment phases

EDUCATION

Postgraduate in Artificial Intelligence and Data Analytics

Saskatchewan Polytechnic, 2025

Postgraduate in Cloud Computing & Blockchain

Saskatchewan Polytechnic, 2024

Bachelor of Science in Computing

University of Greenwich, 2020

Diploma in Software Engineering

National Institute of IT, 2009

CERTIFICATIONS

[Machine Learning, Deep Learning + AWS Sagemaker](#), Udemy - January 2025

[RPA - Process Automation using UIPATH - Beginner to Expert](#), Udemy - April 2022

[Master in Microsoft Power BI Desktop and Service](#), Udemy - March 2022

[Data Warehouse Developer-SQL Server/ETL/SSIS/SSAS/SSRS/T-SQL](#), Udemy - March 2022

AWARDS

Saskatchewan Polytechnic Dean's Honour List (2024)

Outstanding Award for Business Promotion Initiatives (*December 2017*)

Staff of the Month Awards (*2014 - 2019*)